

1. Introduction to Electronics & Safety

1.1 What is Electronics?

Electronics is the branch of science that deals with the flow and control of electrons through devices and circuits. It focuses on low-voltage applications such as computers, communication systems, control systems, and consumer electronics.

1.2 Importance of Electronics in IT & Hardware

- Forms the base of computers and networking devices
- Used in system boards, power supplies, and peripherals
- Helps in troubleshooting and maintenance of IT equipment

1.3 Lab Safety Rules (Very Important)

- Always turn **OFF power** before connecting or removing components
- Never work with **wet hands** or in a damp environment
- Use **insulated tools** only
- Do not short circuit components
- Check voltage and current ratings before use
- Handle ICs, MOSFETs with **anti-static precautions**

1.4 Basic Electronics Kit

- Breadboard (for temporary circuits)
 - DC Power Supply / Battery
 - Connecting wires (jumper wires)
 - Resistors, capacitors, LEDs
 - Switches and push buttons
 - Multimeter (measurement tool)
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2. Passive Components

Passive components do not amplify signals. They only store, limit, or control energy.

2.1 Resistors

Function: Limit current and divide voltage

Unit: Ohm (Ω)

Types:

- Carbon resistor
- Metal film resistor
- Variable resistor (Potentiometer)

Resistor Color Code:

- 1st Band: First digit
- 2nd Band: Second digit
- 3rd Band: Multiplier
- 4th Band: Tolerance

Uses:

- Current limiting (LED protection)
 - Voltage divider circuits
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2.2 Capacitors

Function: Store electrical charge temporarily

Unit: Farad (F)

Types:

- Ceramic capacitor (non-polarized)
- Electrolytic capacitor (polarized)

 Electrolytic capacitors must be connected with correct polarity

Uses:

- Filtering noise in power supply
 - Signal coupling and decoupling
 - Timing circuits
-

2.3 Inductors

Function: Store energy in magnetic field

Unit: Henry (H)

Uses:

- Filters
 - Power supply circuits
 - Noise suppression
-

3. Active Components

Active components require external power and can amplify signals.

3.1 Diodes

Function: Allow current in one direction only

Biasing:

- Forward bias → Conducts
- Reverse bias → Blocks current

Uses:

- Rectifiers
 - Reverse polarity protection
-

3.2 LEDs (Light Emitting Diodes)

- Emit light when forward biased
- Long leg → Anode (+)
- Short leg → Cathode (-)

Uses:

- Indicators
 - Displays
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3.3 Transistors (NPN & PNP)

Function: Switching and amplification

Terminals:

- Emitter
- Base
- Collector

NPN: Turns ON when base is HIGH

PNP: Turns ON when base is LOW

3.4 MOSFETs

Function: Voltage-controlled switching device

Terminals: Gate, Drain, Source

Advantages:

- High speed
 - Low power consumption
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3.5 Integrated Circuits (ICs)

ICs contain thousands to millions of components in a single chip.

Examples:

- Logic ICs (AND, OR, NOT)
 - Timer IC (555)
 - Microcontrollers
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4. Connectors and Electromechanical Components

Switches

- Used to make or break circuits
- Types: Push button, Toggle, Slide

Relays

- Electrically controlled switch
- Used to control high voltage using low voltage

Buzzers

- Produce sound for alerts

Crystals

- Generate stable clock frequency
 - Used in processors and communication circuits
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5. Reading Datasheets & Packaging Types

5.1 Datasheets

Datasheets provide:

- Pin configuration
- Electrical ratings
- Maximum limits
- Application circuits

5.2 Packaging Types

- **DIP:** Easy to use on breadboards
 - **SMD:** Small size, used in modern devices
 - **TO-92:** Small signal transistors
 - **TO-220:** Power devices
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6. Testing and Troubleshooting Components

6.1 Using a Multimeter

- Voltage measurement
- Current measurement
- Resistance measurement
- Continuity testing
- Diode testing

6.2 Troubleshooting Steps

1. Check power supply
 2. Look for loose connections
 3. Test components individually
 4. Replace faulty components
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Final Summary

This detailed unit covers electronics basics, passive and active components, safety rules, identification, packaging, datasheets, and troubleshooting. It forms a strong foundation for hardware maintenance and IT system support.

Buzzers in Electronics

What is a buzzer?

A **buzzer** is an electronic device that **produces sound** when electricity is applied.

In simple words:

A buzzer is used to **make a beep or alarm sound**.

Why buzzers are used

- To give **alerts or warnings**
 - To indicate **errors or notifications**
 - To show **button press confirmation**
-

Types of buzzers

1. Active buzzer

- Makes sound **automatically** when power is given

- No extra signal needed

Example: Doorbell, alarm system

2. Passive buzzer

- Needs a **signal (frequency)** to make sound
- Sound depends on input signal

Example: Musical tone in toys, Arduino projects

Simple real-life examples

- Alarm clock
 - Washing machine beep
 - Mobile phone notification sound
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Key point to remember

A buzzer converts **electrical energy into sound energy**.

Crystals (Crystal Oscillators) in Electronics

What is a crystal?

A **crystal** is an electronic component that **generates a very accurate frequency** using a quartz crystal.

In simple words:

A crystal keeps **time and speed** correct in electronic devices.

Why crystals are used

- To provide a **stable clock signal**
- To control **timing and frequency**
- To synchronize electronic circuits

How a crystal works (simple idea)

- Quartz crystal vibrates when voltage is applied
- This vibration produces a **fixed frequency**

Common crystal frequencies

- **32.768 kHz** – digital clocks
- **8 MHz, 16 MHz** – microcontrollers
- **12 MHz** – communication circuits

Simple real-life examples

- Digital watches
- Computers and mobile phones
- Microcontrollers (Arduino, PIC)

Key point to remember

A crystal provides a **stable and precise timing signal**.

What is a datasheet?

A **datasheet** is a document provided by the manufacturer that explains **everything about an electronic component**.

In simple words:

A datasheet tells us **how to use a component safely and correctly**.

Why datasheets are important

- To know **pin connections**
 - To understand **voltage and current limits**
 - To avoid **damaging components**
 - To design circuits correctly
-

Main parts of a datasheet (easy to remember)

1. General description

- What the component is
- What it is used for

Example: "This IC is a voltage regulator"

2. Pin configuration / Pin diagram

- Shows **pin numbers and pin functions**
- Very important for wiring

Example: VCC, GND, OUTPUT pins

3. Absolute maximum ratings

- Maximum voltage, current, temperature
- **Never exceed these values**

Exam tip: Exceeding these values can permanently damage the component.

4. Recommended operating conditions

- Safe values for normal operation
 - Best performance range
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5. Electrical characteristics

- Current consumption
 - Output voltage
 - Frequency, power rating
-

6. Timing diagrams / graphs (if any)

- Shows how signals behave over time
 - Used in digital components
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7. Application circuits

- Sample circuits provided by manufacturer
- Helps beginners understand usage

Simple real-life example

Before using a **mobile charger**, you check its voltage rating.
Similarly, before using a component, you check its **datasheet**.

Chapter 2: Introduction of Networking Components

2.1 Overview of Network Devices

Network devices are hardware components used to **connect computers and other devices** so that they can share data, resources, and internet connectivity.

1. Modem

A **Modem** (Modulator–Demodulator) is a device that converts digital signals from a computer into analog signals for transmission over communication lines and vice versa.

A modem ([Modulator-Demodulator](#)) is a crucial hardware device that translates digital signals from your computer/router into analog signals (or vice versa) for transmission over physical lines (cable, phone, fiber) to your Internet Service Provider (ISP), enabling internet access by converting data so your devices and the internet can "talk" to each other. It's the bridge between your home network (LAN) and the wider internet (WAN).

How it Works

- [Modulation](#) (Digital to Analog): When you send data (like an email), the modem converts the 0s and 1s (digital) from your devices into analog waves suitable for traveling through cables or phone lines.
- [Demodulation](#) (Analog to Digital): When data comes in from the internet, the modem reverses the process, converting the analog signals back into digital data your computer can understand

Functions:

- Connects a local network to the Internet
- Converts signals for communication

Types:

- DSL Modem
- Cable Modem
- Fiber Modem

These are the most common modems used today for residential and business internet.

- **Cable Modem:** Uses coaxial cables—the same infrastructure as cable TV—to deliver high-speed internet, often reaching speeds up to 100 Mbps or higher.
- **DSL (Digital Subscriber Line):** Connects to the internet via standard copper telephone lines. It is faster than dial-up and allows you to use the phone and internet simultaneously.
- **Fiber Optic Modem:** Converts light signals from fiber optic cables into electronic signals. This is the fastest type available, capable of speeds up to 10 Gbps.
- **Satellite Modem:** Communicates via a satellite dish. It is primarily used in rural or remote areas where cable or DSL infrastructure is unavailable.
- **Dial-up Modem:** The oldest type, using a telephone line to "dial" an ISP. It is much slower than broadband (max 56 kbps) and occupies the phone line during use

Use:

Used at homes and offices to access the Internet.

2. Network Interface Card (NIC)

A **NIC** is a hardware component that allows a computer to connect to a network.

Features:

- Can be wired or wireless

- Has a unique **MAC address**
- Installed internally or externally (USB NIC)

Use:

Without a NIC, a device cannot communicate on a network.

3. Router

A **Router** is a networking device that connects multiple networks and directs data packets between them.

Functions:

- Routes data between devices and the Internet
- Assigns IP addresses (using DHCP)
- Provides security using firewalls

Use:

Commonly used in home and enterprise networks.

4. Switch

A **Switch** is a device used to connect multiple devices within a Local Area Network (LAN).

Features:

- Sends data only to the destination device
- Uses MAC addresses
- Faster and more efficient than a hub

Use:

Used in offices, labs, and data centers.

5. Hub

A **Hub** is a basic networking device that connects multiple computers.

Features:

- Broadcasts data to all connected devices
- Causes more network traffic
- Less secure and slower

Note:

Hubs are mostly obsolete today.

6. Access Point (AP)

An **Access Point** allows wireless devices to connect to a wired network.

Functions:

- Extends Wi-Fi coverage
- Supports multiple wireless users

Use:

Used in schools, offices, malls, and public places.

2.2 Cabling & Connectors

Network cables and connectors provide **physical connectivity** between devices.

1. RJ45 Connector

- Standard connector used for Ethernet cables

- Has 8 pins
- Used with CAT5 and CAT6 cables

Crimping RJ45:

Attaching an RJ45 connector to a cable using a crimping tool.

2. CAT5 and CAT6 Cables

These are twisted-pair Ethernet cables.

Cable Type	Speed	Usage
CAT5	Up to 100 Mbps	Basic LAN
CAT6	Up to 1 Gbps or more	High-speed networks

3. Coaxial Cable

- Single copper conductor
 - Shielded to reduce interference
 - Used in cable TV and older networks
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4. Fiber-Optic Cable

- Uses light signals
- Very high speed and bandwidth
- Immune to electromagnetic interference

Use:

Internet backbone, long-distance communication.

5. Rollover Cable

- Used to connect a computer to a router/switch console port
- Mainly for configuration and management

6. Ethernet Crossover Cable

- Used to connect similar devices directly
- Transmit and receive wires are crossed

Examples:

- PC to PC
- Switch to Switch

7. Straight-Through Cable

- Used to connect different devices
- Most commonly used Ethernet cable

Examples:

- PC to Switch
 - Router to Switch
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2.3 Testing and Troubleshooting Common Physical Issues

Physical layer issues are related to **hardware and cabling problems**.

1. Simulating Network Faults

- Disconnecting cables or devices intentionally
 - Helps understand network behavior during failures
 - Used for training and troubleshooting practice
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2. Replacing Cables and Devices

- Damaged cables cause network failures
 - Faulty NICs, switches, or routers must be replaced
 - Visual inspection is often the first step
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3. Cable Tester

A **Cable Tester** is used to check the correctness of network cables.

Checks:

- Continuity
- Wire mapping
- Open or short circuits
- Crossed pairs

Tools & Equipment Needed

These tools are essential for **network setup and maintenance**.

a) Wireless Router, Ethernet Switch, Hub

- Used to build and test networks

b) RJ45 Cables (CAT5/6), Crimping Tools

- Used to create custom Ethernet cables

c) LAN Tester, Multimeter

- **LAN Tester:** Tests Ethernet cables
- **Multimeter:** Measures voltage, resistance, and continuity

d) Access Points, Extenders

- Improve wireless signal coverage
- Extend network range

3. Desktop and Laptop Hardware Identification and Handling (6 Hours)

3.1 Introduction to PC Hardware

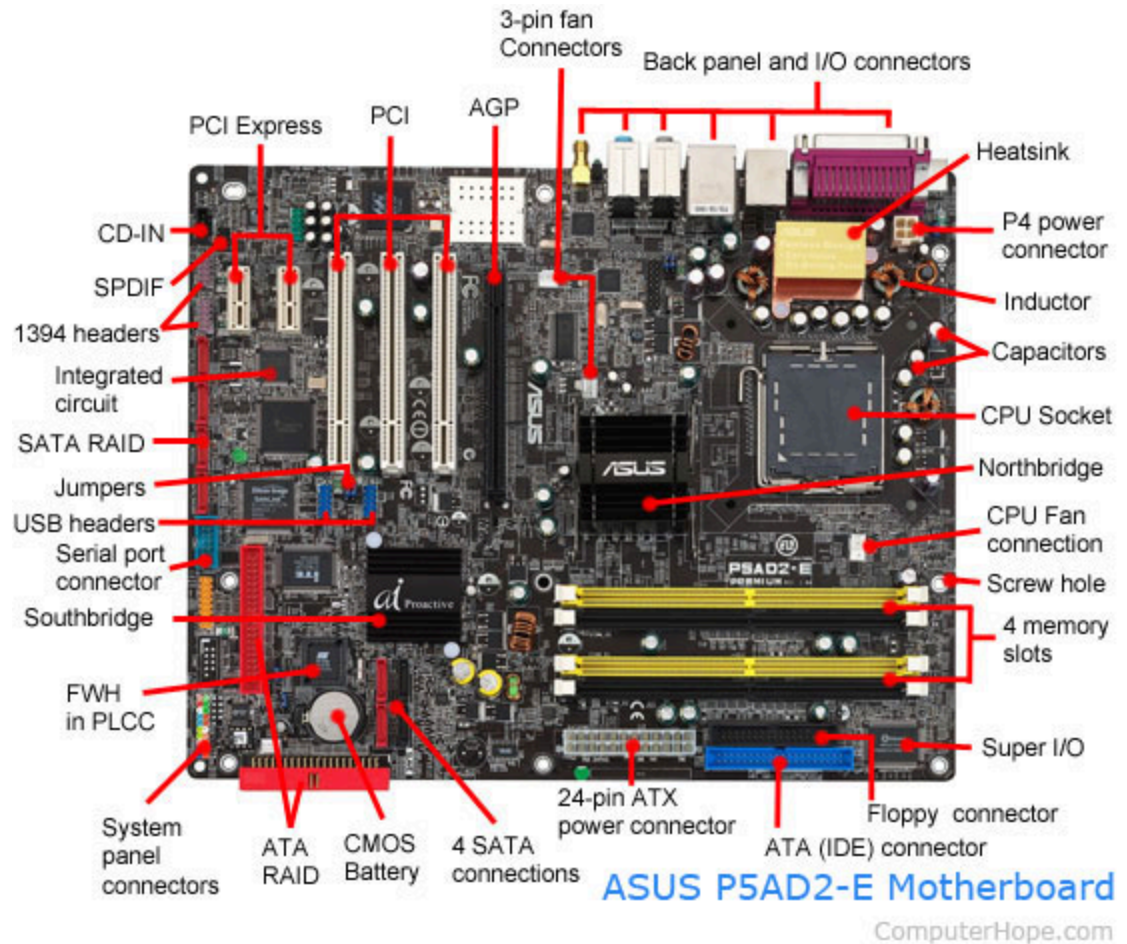
A. Basic Computer Components

1. Case (Cabinet)

- The outer enclosure that houses all internal computer components.
- Protects components from dust, physical damage, and electromagnetic interference.
- Provides airflow and mounting space for hardware.

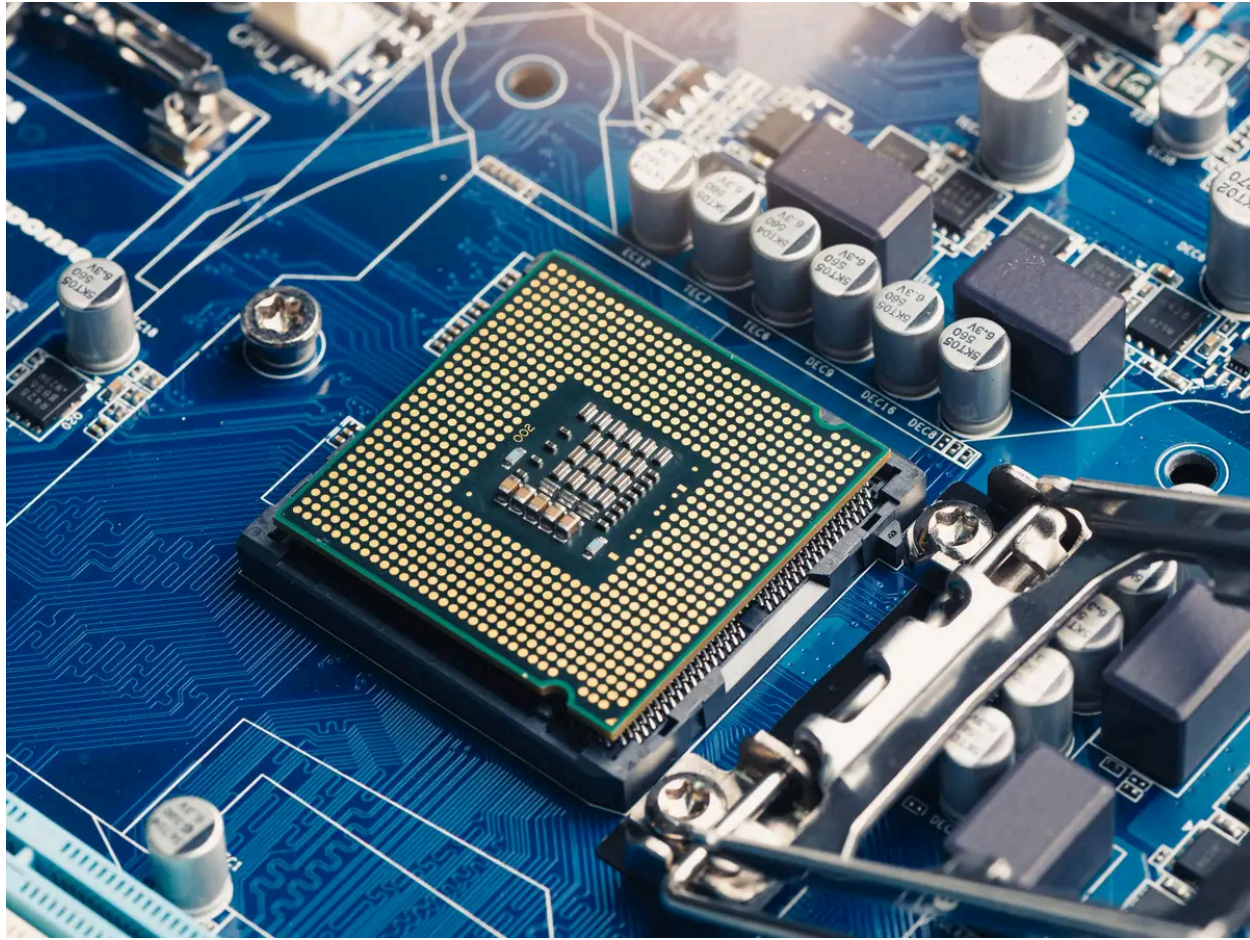
2. Motherboard

- The main circuit board of the computer.
- Connects and allows communication between all components like CPU, RAM, storage, and peripherals.
- Contains chipset, slots, ports, and connectors.



3. CPU (Central Processing Unit)

- Known as the “brain” of the computer.
- Performs calculations, executes instructions, and processes data.
- Speed measured in GHz.



4. RAM (Random Access Memory)

- Temporary (volatile) memory used to store data currently in use.
- Faster than storage devices.
- Data is lost when power is turned off.



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5. ROM (Read Only Memory)

- Non-volatile memory that stores permanent instructions.
- Contains firmware like BIOS/UEFI.
- Retains data even when power is off.

6. PSU (Power Supply Unit)

- Converts AC power from the wall socket into DC power for internal components.
- Supplies different voltages required by hardware.

B. Types of Components

1. Types of Computer Cases

- **Desktop Case** – Horizontal, used in offices.
- **Tower Case**
 - *Mini Tower*
 - *Mid Tower*
 - *Full Tower*
- **Laptop Case** – Compact, integrated design.

2. Types of Motherboards

- **ATX** – Standard size, more expansion slots.
- **Micro-ATX** – Smaller, fewer slots.
- **Mini-ITX** – Very small, compact systems.

3. Types of RAM

- **SDRAM** – Older type, synchronized with system clock.
- **DDR (Double Data Rate)**
 - **DDR**
 - **DDR2**
 - **DDR3**
 - **DDR4**
- Each newer version is faster and more power-efficient.
- Different DDR types are **not compatible** with each other.

4. Types of ROM

- **PROM (Programmable ROM)** – Can be programmed once.
- **EPROM (Erasable Programmable ROM)** – Erased using UV light.
- **EEPROM (Electrically Erasable PROM)** – Erased electrically (used in BIOS).

5. Types of Power Supply Units

- **AT PSU** – Older type.
- **ATX PSU** – Modern standard.
- **Modular PSU** – Detachable cables.

- **Non-Modular PSU** – Fixed cables.
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C. Front Panel and Back Panel Components

Front Panel Components

- Power button
- Reset button
- USB ports
- Audio ports (Headphone/Mic)
- Power LED and HDD activity LED
- Optical drive (older systems)

Back Panel Components

- USB ports
 - HDMI / VGA / DisplayPort
 - Ethernet (LAN) port
 - Audio jacks
 - PS/2 ports (keyboard/mouse – older)
 - Power socket
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3.2 Preparing the Workbench & Safety

A. ESD (Electrostatic Discharge) Safety

- Static electricity can damage electronic components.
- Precautions:
 - Use anti-static wrist strap.
 - Work on anti-static mat.
 - Avoid touching pins and circuits.
 - Keep components in anti-static bags.

B. Tool Handling

- Use correct tools:
 - Screwdrivers (Phillips, flathead)
 - Tweezers
 - Cable ties
- Handle tools carefully to avoid damaging components.
- Never use excessive force.

C. Workspace Preparation

- Clean, dry, and well-lit area.
- No liquids near components.
- Organized layout to avoid losing screws.
- Proper ventilation.

3.3 System Unit Components

A. Internal Components

1. **Motherboard**

- Holds and connects all components.
- Contains chipset, slots, and ports.

2. **CPU**

- Executes instructions and calculations.
- Installed in CPU socket.

3. **Northbridge**

- Handles high-speed communication.
- Connects CPU with RAM and graphics.
- Mostly integrated into CPU in modern systems.

4. **Southbridge**

- Handles low-speed devices.
- Controls USB, hard drives, audio, and I/O devices.

5. **RAM**

- Temporary memory for running programs.
- Faster access than storage.

6. **Power Supply**

- Provides electrical power to all components.
- Includes cooling fan.

7. **BIOS / UEFI**

- Firmware stored in ROM.
- Initializes hardware during startup.

- Loads operating system.
 - UEFI is modern replacement for BIOS.
-

B. Internal Buses and Slots

Internal Buses

- **Data Bus** – Transfers data.
- **Address Bus** – Identifies memory locations.
- **Control Bus** – Carries control signals.

Expansion Slots

- **PCI** – Older expansion cards.
 - **PCI Express (PCIe)** – Modern high-speed slot.
 - **RAM Slots (DIMM)** – For memory modules.
 - **M.2 Slot** – For SSDs and Wi-Fi cards.
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C. Identifying Components in a System

- Locate and label:
 - CPU
 - RAM slots
 - Power connectors
 - Expansion slots
 - Storage connectors

- Understand **function of each component**:

- CPU → processing
- RAM → temporary memory
- PSU → power supply
- Motherboard → communication hub

A computer system is made up of different hardware components. Each component has a specific function that helps the computer work properly.

3.4 Storage Devices

What is Storage?

Storage devices are used to **store data permanently**, even when the computer is turned off.

Types of Storage Devices

1. Hard Disk Drive (HDD)

- Uses **spinning magnetic disks**
- Slower than SSD
- Cheaper
- Larger storage capacity
- Common in older computers

2. Solid State Drive (SSD)

- No moving parts
- Much **faster than HDD**
- More durable

- Uses flash memory
- More expensive than HDD

3. Optical Drives

- Use **CD, DVD, or Blu-ray discs**
 - Used for reading/writing discs
 - Becoming less common
-

Storage Interfaces

Interfaces are how storage devices connect to the motherboard.

- **PATA (IDE)** – Old technology, slow, not used now
 - **SATA** – Common for HDDs and SSDs
 - **NVMe** – Very fast interface used with modern SSDs
 - **SSD (Form factor)** – Can be SATA or NVMe
-

Removing & Reinstalling a Drive

- Power off the computer
- Disconnect cables
- Remove the drive carefully
- Reinstall securely and reconnect cables

Real vs Fake SSD

- Real SSD: Fast boot time, proper brand, correct storage size
 - Fake SSD: Very slow, fake capacity, unreliable performance
-

3.5 Input / Output Devices

Input Devices

Used to send data **into** the computer:

- Keyboard
- Mouse
- Scanner

Output Devices

Used to show results **from** the computer:

- Monitor
- Printer

(Some devices like touch screens work as both input and output.)

Connecting and Testing Devices

- Plug devices into correct ports
- Install drivers if needed
- Test functionality in system settings

Common Port Types

- **USB** – Keyboard, mouse, printer, storage devices
 - **HDMI** – Monitor, projector
 - **VGA / DisplayPort** – Display devices
 - **Audio Jack** – Headphones, microphone
-

3.6 Power Supply Unit (PSU)

Role of PSU

- Converts AC power to DC power
 - Supplies power to all components
 - Protects components from voltage issues
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3.6.1 PSU Overview

ATX PSU

- Standard size for desktop computers

Modular PSU

- Detachable cables
- Better cable management
- Improved airflow

SMPS (Switched Mode Power Supply)

- Efficient power conversion

- Used in modern computers
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PSU Models and Internal Parts

Internal components include:

- Transformer
- Capacitors
- Cooling fan
- Rectifier
- Voltage regulator

Each part helps in safe and stable power delivery.

Testing the PSU

- Use a PSU tester or multimeter
- Check if the fan spins
- Ensure correct voltage output

What they are

- **HDD (Hard Disk Drive):** Uses spinning disks (platters) and a moving read/write arm.
- **SSD (Solid State Drive):** Uses flash memory (no moving parts).

2. Speed

- **HDD:** Slower. Booting an OS or opening apps can take noticeable time.

- **SSD:** Much faster. Computers start up and load programs quickly.

3. Durability

- **HDD:** More fragile because of moving parts; drops can damage it.
- **SSD:** More durable and shock-resistant.

4. Noise & heat

- **HDD:** Can make noise and generate more heat.
- **SSD:** Silent and cooler.

5. Power usage

- **HDD:** Uses more power.
- **SSD:** More energy-efficient (better for laptops).

6. Storage capacity & price

- **HDD:** Cheaper per gigabyte; good for very large storage (movies, backups).
- **SSD:** More expensive per gigabyte, but prices are dropping.

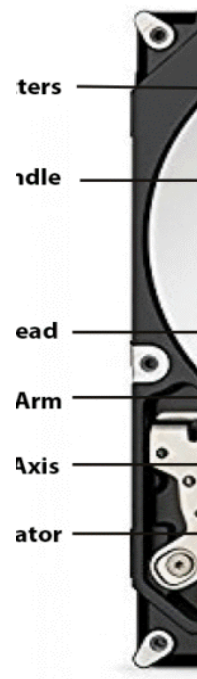
7. Typical use cases

- **HDD:** Long-term storage, backups, large files.
- **SSD:** Operating system, games, apps, and anything that benefits from speed.

Quick summary

Feature	HDD	SSD
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Speed	Slow	Very fast
Moving parts	Yes	No
Durability	Lower	Higher
Noise	Yes	No
Cost per GB	Cheaper	More expensive



4.1 Safety Precautions

Before starting PC or laptop assembly, the following safety precautions must be followed:

- Switch off and unplug the power supply
 - Use an anti-static wrist strap or ensure proper grounding
 - Avoid touching component pins and circuitry
 - Handle components carefully by their edges.
 - Keep liquids away from electronic components.
 - Ensure capacitors are discharged before handling power components
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4.2 Step-by-Step Assembly of a PC

Hands-on Group Activity

- Identify and verify all components
 - Assemble components systematically
 - Power on the system and confirm successful boot to BIOS
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4.3 Installing Motherboard, CPU, and RAM

Motherboard Installation

- Fix spacers (standoffs) inside the cabinet
- Align the motherboard with the I/O shield
- Secure the motherboard using screws

CPU Installation

- Open the CPU socket lever
- Align the CPU correctly using the orientation mark
- Place the CPU gently into the socket and lock it

Applying Thermal Paste

- Apply a small amount of thermal paste on the CPU surface.
- Ensures efficient heat transfer to the heat sink.

RAM Installation

- Insert RAM modules into the correct DIMM slots
 - Press firmly until the locking clips snap into place
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4.4 Connecting PSU, Storage, and Cables

Power Supply Unit (PSU) Connections

- Connect the 24-pin ATX power cable to the motherboard
- Connect the CPU power cable (4-pin or 8-pin)

Storage Connections

- Connect SATA data cables from storage device to motherboard
- Connect SATA power cables from PSU to storage devices

Front Panel Connections

- Power switch
 - Reset switch
 - Hard disk LED
 - Front USB and audio connectors
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4.5 BIOS / UEFI Configuration and POST Checks

Accessing BIOS/UEFI

- Press **DEL**, **F2**, **F10**, or **ESC** during system startup

BIOS/UEFI Configuration

- Set boot device priority
- Configure system date and time
- Enable or disable hardware features
- Set BIOS/UEFI passwords

POST (Power-On Self-Test)

- Checks CPU, RAM, storage, and peripherals
 - Errors indicated through:
 - Beep codes
 - On-screen error messages
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4.6 Troubleshooting Basics

POST and Beep Codes

- Repeated beeps: RAM-related issues
- No display: GPU or RAM fault
- No power: PSU or motherboard problem

Common Hardware Issues

- Loose cables
- Improperly seated components
- Faulty hardware parts

Diagnostic Tools

- **Multimeter** – Measures voltage and continuity
 - **POST card** – Displays motherboard error codes
 - Hardware fault simulation for practice and learning
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4.7 Installing Operating System (Optional)

Operating System Installation

- Boot system using USB or DVD
- Install Windows or Linux OS

Post-Installation Tasks

- BIOS and firmware updates
- Device driver installation

- Disk cleanup and optimization
- Antivirus installation and system scan

Peripheral Installation

- Virtual printer setup
 - Physical printer installation
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4.8 Useful Operating System Commands

Smart and Useful Commands

- File and directory management
- Process and system monitoring
- Network troubleshooting commands

Cross-Platform Commands

- Common commands used in both Windows and Linux environments
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4.9 Troubleshooting and Diagnostics

- Identify hardware and software faults
- Use diagnostic utilities and system logs
- Apply systematic troubleshooting techniques to resolve issues

6. Basic Networking Configuration (2 Hours)

6.1 IP Addressing & Subnetting

1. IP Address

An IP (Internet Protocol) address uniquely identifies a device on a network. It consists of:

- **Network ID** – Identifies the network
- **Host ID** – Identifies the device within the network

Example: 192.168.1.10

0.0.0.0–

255.255.255.255

2. IP Classes

IPv4 addresses are divided into classes:

Classes	Range	Default Subnet Mask	Usage
A	1–126	255.0.0.0	Large networks
B	128–191	255.255.0.0	Medium networks
C	192–223	255.255.255.0	Small networks

3. Subnet Mask

A subnet mask separates the network portion and host portion of an IP address.

Example:

- IP: 192.168.1.10

- Subnet Mask: 255.255.255.0

4. Gateway

A **default gateway** is the router IP address that connects a local network to other networks (like the Internet).

5. DNS (Domain Name System)

DNS translates domain names (e.g., google.com) into IP addresses.

6.2 Router & Switch Configuration

Router Configuration

- Assign IP address
- Configure subnet mask
- Set default gateway
- Enable interfaces
- Save configuration

Switch Configuration

- Assign management IP
- Configure VLAN (if required)
- Test connectivity

Connectivity Testing

- Use **ping** to test communication between devices.
- Verify network status.

Console Setup

- Connect router/switch to PC using console cable.
 - Use terminal software (e.g., PuTTY).
 - Configure device via CLI.
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6.3 Wireless Access Points (AP) & Extenders

Setup Process:

1. Configure **SSID** (Wi-Fi name).
2. Set a secure **password** (WPA2/WPA3 recommended).
3. Enable **MAC Filtering** (optional for extra security).
4. Perform **signal strength test**.

Extenders

- Used to increase Wi-Fi coverage area.
 - Should be placed between router and weak signal area.
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6.4 Network Design & Topologies

Common Network Topologies:

1. **Star Topology**
 - All devices connect to a central switch/hub.
 - Easy to manage and troubleshoot.

2. Bus Topology

- All devices share a single communication line.
- Less expensive but difficult to troubleshoot.

Practical Task

- Build star and bus topology using real devices.
 - Test communication between connected systems.
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6.5 Testing & Diagnostics

Important Networking Commands:

Command	Purpose
<code>ping</code>	Tests connectivity
<code>tracert</code>	Shows path to destination
<code>ipconfig</code>	Displays IP configuration
<code>netstat</code>	Shows active connections
<code>arp</code>	Displays ARP table
<code>nslookup</code>	Tests DNS resolution
<code>route</code>	Shows routing table
<code>netsh</code>	Network configuration tool

Cable Testers

- Used to check LAN cable faults.
- Detects open, short, or miswired cables.

6.6 Troubleshooting Common Issues

Common Problems:

- No internet connection
- IP conflict
- Loose/damaged cable
- Incorrect subnet mask
- Weak Wi-Fi signal

Troubleshooting Steps:

1. Check physical connections.
2. Verify IP configuration.
3. Ping gateway.
4. Restart router/switch.
5. Replace faulty cables/devices.

Practical Activity

Mini LAN/WLAN Setup Task

- Create a small LAN using switch and PCs.
- Configure IP addresses manually.
- Setup Wi-Fi using AP.

- Test connectivity.
- Demonstrate working setup (Demo & Viva).

7. Problem Solving Techniques (Estimated Time: 2 Hours)

Effective problem solving is an essential skill in IT support, computer maintenance, and networking. A structured approach helps technicians diagnose issues accurately, reduce downtime, and prevent repeated problems.

7.1 Start with Visual Inspection

Before using tools or running commands, always begin with a physical and visual check.

Why?

Many technical problems are caused by simple physical issues that can be fixed quickly.

What to Check:

- Loose or unplugged cables (power, HDMI, Ethernet, USB)
- Damaged ports (bent pins, broken connectors)
- Power supply connections
- Overheating signs (dust buildup, blocked vents)
- Physical damage (burn marks, cracked components)
- Proper seating of RAM, GPU, or expansion cards

Example:

If a computer monitor shows “No Signal,” first check:

- Is the monitor powered on?
- Is the HDMI/VGA cable connected properly?
- Is the cable damaged?

Visual inspection can save time and prevent unnecessary advanced troubleshooting.

7.2 Use the Elimination Method (Test One Part at a Time)

This method involves isolating and testing components individually to find the faulty one.

Why?

Testing everything at once can cause confusion. Changing one variable at a time helps identify the real cause.

How to Apply:

1. Identify possible causes.
2. Test one component.
3. Observe results.
4. Move to the next component if needed.

Example:

If a computer is not booting:

- Test with a different power cable.
- Then test with another RAM stick.
- Then test another power supply.
- Then check the motherboard.

By eliminating working components, you narrow down the problem.

7.3 Check BIOS/UEFI Messages and Beep Codes

When a computer starts, BIOS/UEFI performs a Power-On Self-Test (POST).

Why?

Error messages or beep codes give direct clues about hardware problems.

What to Look For:

- Error messages on screen (e.g., “No Boot Device Found”)
- Continuous or repeated beep sounds
- Boot failure messages

Common Beep Code Issues:

- Continuous beeps → RAM issue
- No beep → Power supply or motherboard issue
- Multiple short beeps → Hardware failure

Always refer to the motherboard manual for exact beep code meanings.

7.4 Use Diagnostic Tools for Network Issues

For network troubleshooting, software tools help identify connection and configuration problems.

Important Tools:

1. Ping

Tests connectivity between two devices.

- Example: `ping 8.8.8.8`
- If reply received → Network connection is working.
- If request timed out → Possible network or firewall issue.

2. ipconfig (Windows) / ifconfig (Linux/Mac)

Displays IP configuration details.

- Check IP address
- Check default gateway
- Check subnet mask

Example:

If IP starts with 169.254.x.x → Device failed to obtain IP from DHCP.

3. Network Testers

- Cable testers (check Ethernet cables)
- Loopback plugs
- Network analyzers

These tools help determine if the problem is:

- Cable related
- Router related
- ISP related
- Device configuration related

7.5 Maintain a Troubleshooting Log

Documentation is extremely important in technical work.

Why?

- Helps track recurring problems.
- Saves time in future troubleshooting.
- Provides record for reporting and audits.
- Improves learning and experience.

What to Include:

- Date and time
- Problem description
- Steps taken
- Tools used
- Final solution
- Preventive actions

Example Entry:

Date: 10 Feb 2026
Issue: No internet connection
Cause: Faulty Ethernet cable
Solution: Replaced cable

Keeping records improves efficiency and professionalism.

7.6 Follow a Systematic Approach

A structured process prevents guesswork and ensures reliable results.

Standard Troubleshooting Flow:

1. Identify

- Gather information from the user.
- Observe symptoms.
- Ask questions.
- Check error messages.

2. Test

- Perform basic checks.
- Use tools to confirm the issue.

3. Isolate

- Narrow down the root cause.
- Remove possible causes one by one.

4. Fix

- Repair or replace faulty component.
- Update software if required.
- Reconfigure settings.

5. Verify

- Restart system.
- Test functionality.
- Confirm with the user.
- Monitor for recurring issues.

Key Takeaways for Students

- Always start simple.
- Do not guess — test logically.
- Change only one thing at a time.
- Use tools properly.
- Document everything.
- Follow a structured approach every time.

Practical Activity Suggestion (For 2-Hour Session)

1. Demonstrate a loose cable problem.
2. Simulate a RAM error (show beep codes).
3. Practice using:
 - ping
 - ipconfig
4. Create a sample troubleshooting log.
5. Solve a mock network issue using elimination method.

These techniques will help students become confident and systematic IT technicians.

8. Maintenance & Upgrade (5 Hours)

Maintenance and hardware upgrades are essential to keep a computer system running efficiently, securely, and reliably. Proper planning and compatibility checks prevent hardware failure and unnecessary expenses.

A systematic approach should always be followed:

Identify → Test → Isolate → Fix → Verify

8.1 Hardware Upgrade Tips

Hardware upgrades improve system performance, storage capacity, and multitasking ability. However, upgrading requires careful compatibility checks and planning.

8.1.1 Upgrade Planning & Compatibility Checks

Before upgrading any component, ensure it is compatible with your system.

1. Motherboard Compatibility

The motherboard determines which components can be installed.

Check the following:

- **CPU Compatibility**
 - Socket type (e.g., LGA, AM4, AM5)
 - Supported processor generation
 - Chipset support
 - BIOS version requirement
- **RAM Compatibility**

- Supported RAM type (DDR3, DDR4, DDR5)
- Maximum RAM capacity
- Maximum supported speed (MHz)
- Number of RAM slots
- **GPU Compatibility**
 - PCIe slot type (usually PCIe x16)
 - PCIe version (3.0, 4.0, 5.0)
 - Physical space inside the cabinet
 - Power supply requirements

Always check the motherboard manual or official website specifications.

2. Laptop Upgrade Limitations

Laptop upgrades are often limited compared to desktops.

Check:

- Maximum supported RAM
- Number of RAM slots (1 or 2)
- Storage type (SATA, NVMe)
- Number of storage bays
- Whether RAM is soldered (non-removable)
- Whether CPU/GPU is integrated (non-upgradable)

Many modern laptops allow only:

- RAM upgrade (if not soldered)
 - Storage upgrade (SSD replacement)
-

3. Use Manufacturer Documentation & Tools

Always rely on official documentation and system scanning tools.

Useful tools:

- System information utilities (to check RAM type, CPU, motherboard model)
- Memory compatibility scanners
- Manufacturer support websites

These tools help identify:

- Current hardware configuration
 - Maximum upgrade capacity
 - Compatible components
-

8.1.2 Memory (RAM) Upgrades

RAM upgrades significantly improve multitasking, system speed, and application performance.

1. Match RAM Specifications

Before buying RAM, ensure it matches:

- **Type** (DDR3, DDR4, DDR5)
- **Speed** (e.g., 2400 MHz, 3200 MHz)

- **Voltage**
- **Capacity supported by motherboard**
- **Form Factor**
 - SO-DIMM → Laptops
 - DIMM → Desktops

Important:

- Mixing different RAM speeds may cause the system to run at the lowest common speed.
 - Mixing different brands may sometimes cause instability.
-

2. Dual-Channel RAM Installation

Dual-channel mode improves memory bandwidth and performance.

To enable dual-channel:

- Install identical RAM sticks (same size and speed)
- Insert in recommended slots (check motherboard manual)

Example:

- 2 × 8GB performs better than 1 × 16GB

Benefits:

- Better gaming performance
 - Improved multitasking
 - Faster data processing
-

3. BIOS/UEFI Settings for RAM Recognition

After installing new RAM:

1. Enter BIOS/UEFI setup during boot.
2. Check:
 - Total memory detected
 - Memory speed
3. Enable memory profile (XMP/DOCP if available) for rated speed.
4. Save and restart.

If RAM is not detected:

- Reseat RAM module.
 - Update BIOS if required.
 - Check compatibility again.
-

Importance of Proper Upgrade Planning

- Prevents hardware damage
- Saves money
- Avoids compatibility issues
- Ensures stable system performance
- Extends system lifespan

8.1.3 Storage Upgrades

Upgrading storage improves system speed, boot time, application loading time, and overall performance. Choosing the right storage type and installing it correctly is very important.

1. Comparison: SSD vs HDD vs NVMe

(A) HDD (Hard Disk Drive)

- Uses mechanical spinning disks.
- Slower read/write speeds.
- More prone to physical damage.
- Produces noise and heat.
- Cheapest storage option.
- Boot time: Slow (30–60 seconds or more).

Best for: Large storage at low cost (backup, media storage).

(B) SATA SSD (Solid State Drive)

- No moving parts.
- Much faster than HDD.
- More durable and silent.
- Lower power consumption.
- Connected via SATA interface.
- Boot time: 10–20 seconds.

Best for: OS installation and everyday use.

(C) NVMe SSD

- Uses PCIe interface.
- Extremely high speed.
- Much faster than SATA SSD.
- Lower latency.
- Small form factor (M.2).
- Boot time: 5–10 seconds.

Best for: Gaming, video editing, heavy applications.

Speed Comparison (Approximate)

Storage Type	Speed	Durability	Boot Time	Cost
HDD	Slow (100–150 MB/s)	Medium	Slow	Low
SATA SSD	Fast (500–550 MB/s)	High	Fast	Medium
NVMe SSD	Very Fast (2000–7000+ MB/s)	Very High	Very Fast	Higher

2. Use Cloning Tools for OS/Data Migration

When upgrading storage, you can:

Option 1: Clone Old Drive

- Copies OS, files, and settings.
- Saves time.
- No need to reinstall software.

Steps:

1. Connect new drive.
2. Use disk cloning software.
3. Clone old drive to new drive.
4. Change boot priority in BIOS.

Option 2: Clean Installation

- Install fresh operating system.
 - Better system performance.
 - Removes unwanted files and malware.
-

3. Align Partitions Properly for SSDs

Proper partition alignment ensures:

- Maximum speed
- Longer SSD lifespan
- Reduced write amplification

Important:

- Modern operating systems automatically align partitions.

- Avoid defragmenting SSDs.
 - Enable TRIM support for SSD performance.
-

8.1.4 Graphics Card (GPU) Upgrade (Desktop)

GPU upgrades improve gaming, graphic design, 3D rendering, and video editing performance.

1. Check PSU Wattage and Power Connectors

Before upgrading:

- Verify total PSU wattage.
- Check 6-pin / 8-pin power connectors.
- Ensure PSU is from a reliable brand.
- Check GPU recommended power requirement.

If PSU is insufficient:

- Upgrade PSU before installing GPU.
-

2. Check Physical Space Inside the Case

Modern GPUs are large.

Check:

- GPU length

- GPU thickness (2-slot, 3-slot)
- Cabinet size
- Airflow clearance

Ensure:

- No obstruction from hard drives.
 - Enough airflow for cooling.
-

3. Verify PCIe Slots and Compatibility

- Most GPUs use PCIe x16 slot.
 - PCIe versions are backward compatible.
 - Ensure motherboard has required slot.
 - Avoid using damaged or loose PCIe slots.
-

4. Cooling Requirements

High-performance GPUs generate heat.

Ensure:

- Proper case ventilation.
- Enough case fans.
- Clean dust filters.
- Good airflow direction.

Overheating can cause:

- Performance throttling
 - System crashes
 - Hardware damage
-

5. Install Latest Drivers After Upgrade

After installing GPU:

1. Boot system.
2. Install latest graphics drivers.
3. Restart computer.
4. Test performance using benchmark tools.

If switching GPU brands:

- Uninstall old drivers first.
 - Then install new drivers.
-

8.1.5 CPU Upgrade Tips (Desktop)

CPU upgrade improves overall processing speed and multitasking ability.

1. Check Socket Type

CPU must match motherboard socket.

Examples:

- Intel LGA sockets
- AMD AM series sockets

If socket does not match:

- CPU will not fit.
-

2. Check Chipset Support

Even if socket matches:

- Chipset must support that CPU generation.
 - Check motherboard CPU support list.
-

3. BIOS Compatibility

Sometimes:

- BIOS update is required.
- Update BIOS before installing new CPU.

Warning:

- Do not interrupt BIOS update.
 - Power failure may damage motherboard.
-

4. Apply Thermal Paste Correctly

Thermal paste helps transfer heat from CPU to cooler.

Steps:

1. Clean old paste.
2. Apply pea-sized amount in center.
3. Mount cooler evenly.
4. Tighten screws diagonally.

Too much paste can:

- Cause overheating.
 - Reduce cooling efficiency.
-

5. Check Cooling System

High-performance CPUs require better cooling.

Options:

- Stock air cooler
- Advanced air cooler
- Liquid cooling system

Monitor:

- CPU temperature
 - Fan speed
 - System stability
-

6. Test with Stress Tools

After installation:

Run stress testing software to check stability.

Monitor:

- Temperature
- System crashes
- Performance consistency

If temperature is too high:

- Reapply thermal paste.
- Improve cooling.
- Check fan installation.

Final Best Practices for Storage, GPU, and CPU Upgrades

- ✓ Turn off and unplug system
- ✓ Use anti-static precautions
- ✓ Handle components carefully
- ✓ Double-check compatibility
- ✓ Update drivers and BIOS if required
- ✓ Test system after upgrade
- ✓ Monitor temperature and performance